the model's behavior, we focus on two datasets: $\mathcal{D}_p$, a larger dataset representing pre-training or the primary objective of training, and $\mathcal{D}_a$, a smaller dataset representing the alignment process or the secondary objectives of training. Since the pre-training stage encompasses a wide range of data, without loss of generality, we assume that the datasets in the alignment process follow the same distribution as some subset of the pre-training data.

Definition 4.1 (Normalized Compression Rate)**.** For N distinct datasets $\mathcal{D}_1, \cdots, \mathcal{D}_N$ and a parameter model $p_{\boldsymbol{\theta}}$ compressing $\mathcal{D} = \bigcup_{i=1}^{N} \mathcal{D}_i$, the normalized compression rate $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i/\mathcal{D}}$ for a particular dataset $\mathcal{D}_i$ is defined as:

$$\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i/\mathcal{D}} = \gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i} - \log M, \tag{1}$$

where M is the number of leaf nodes of the pruned tree $\mathcal{T}_i'$ of dataset $\mathcal{D}_i$.

The normalized compression rate allows for comparing the model's compression performance across different datasets. The smaller the normalized compression rate of a dataset, the better the model's compression performance on the dataset.

With this definition in hand, we proceed to our main result: language models exhibit *elasticity*.

Theorem 4.2 (*Elasticity* of Language Models)**.** *Consider the pre-training dataset $\mathcal{D}_p = \bigcup_{i=1}^{3} \mathcal{D}_i$, the alignment dataset $\mathcal{D}_a$, and the perturbation dataset $\mathcal{D}_t$, with the model $p_{\boldsymbol{\theta}}(\cdot)$ trained on $\mathcal{D} = \mathcal{D}_p \cup \mathcal{D}_a \cup \mathcal{D}_t$. Assume that $\mathcal{D}_a \overset{d}{\sim} \mathcal{D}_2$, $\mathcal{D}_t \overset{d}{\sim} \mathcal{D}_3$, and $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$ are each distributed according to a Pareto mass distribution (Newman, 2005). As the perturbation data volume $|\mathcal{D}_t|$ varies, $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}$ and $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}$ satisfy:*

$$\begin{aligned} &\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l} = \Theta\left(-k\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}\right), \\ &\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l} < 0, \frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l} > 0, \end{aligned} \tag{2}$$

where $l = \frac{|\mathcal{D}_t|}{|\mathcal{D}_a|} \ll 1$, $k = \frac{|\mathcal{D}_p|}{|\mathcal{D}_a|} \gg 1$, and $\{\mathcal{D}_i\}_{i=1}^{3}$ are datasets of equal cardinality.

Theorem 4.2 shows that as the perturbation increases, the normalized compression rates of the model for $\mathcal{D}_p$ decrease and $\mathcal{D}_a$ increase and the rate of changes is strongly correlated with the size of the datasets. Unlike the proportional changes in compression rates across different datasets, the language model seems to *prefer* the dataset with a larger volume, leading to biased model behavior after the perturbation.

4.2 *Elasticity* and *Inverse Alignment*

Theorem 4.2 shows the normalized compression rate change of different datasets is inversely proportional to their sizes under perturbations. A significant size difference between the pre-training and alignment datasets causes rapid performance degradation on the alignment dataset due to the inverse relationship, often spanning several orders of magnitude.

Intuitively, the model's compression process across multiple datasets is akin to resource allocation between towns: in a region with both a large metropolis and rural villages, to maximize overall economic productivity, resources are typically allocated to the metropolis first, to leverage its scale and agglomeration effects. In other words, the metropolis/large dataset occupies a dominant position in the system.

Therefore, the *elasticity* of language models makes *inverse alignment* possible. Due to the substantial data volume disparity between pre-training and alignment datasets, the model tends to revert to the pre-training rather than alignment distribution when subsequent perturbations occur. This indicates an inherent tendency toward *inverse alignment* during fine-tuning. Maximizing the impact of *elasticity* through well-designed perturbations will be key to achieving true *inverse alignment*.

4.3 *Elasticity* and the Hooke's Law.

The inverse proportionality result in Theorem 4.2 provides a potential invariant in the model training and alignment process: after perturbation, the rate of change in the compression rates of different datasets is inversely proportional to their sizes, with the absolute value of the product being a constant. This constant characterizes the impact of the perturbation on the model and indirectly describes the model's resistance to perturbations, or its *elasticity*.

The *elasticity* of the model can be intuitively analogized to a series system of springs, as shown in Figure 1. Consider two massless springs in series, with spring constants k_1 and k_2, respectively. When the entire system undergoes deformation due to an external force F, the system reaches a stable state, and the elastic force exerted by each spring is equal to F. According to Hooke's Law (Hooke, 2016), the elongation Δl_1 and Δl_2 of each spring is

inversely proportional to its spring constant. Thus, in this system, we have:

$$F \propto k_1 \cdot \Delta l_1 = k_2 \cdot \Delta l_2 \ .$$

In the language model setting, after integrating Theorem 4.2 to l, we obtain $\Delta\gamma_{p_\theta}^{\mathcal{D}_i/\mathcal{D}}$ across different datasets, which is equivalent to the change in the KL divergence $\Delta D_{\mathrm{KL}}(\mathcal{P}_{p_\theta}||\mathcal{P}_{\mathcal{D}_i})$ between the model's distribution and the distributions of the individual datasets, is inversely proportional to the size of the datasets $|\mathcal{D}_i|$. Here, we only consider the absolute value of ΔD_{KL}. Analogous to the series spring model, the *elasticity* F in LLMs satisfies:

$$F \propto |\mathcal{D}_i| \cdot \Delta D_{\mathrm{KL}}(\mathcal{P}_{p_\theta}||\mathcal{P}_{\mathcal{D}_i}) \ , \tag{3}$$

where ΔD_{KL} corresponds to Δl in the spring model, while $|\mathcal{D}|$ corresponds to the spring constant k, thus leading to the *elasticity* of LLMs.

5 How *Elasticity* Resists Alignment?

In the previous sections, we proved that LLMs have *elasticity*. This section will analyze two specific phenomenons of it:

- ***Resistance* for Pre-Trained Models.** Models tend to maintain the original distribution and resist alignment;
- ***Rebound* for Post-Trained Models.** Fine-tuning in the opposite direction of post-training (*e.g.*, safe *vs.* unsafe) causes post-trained models to return quickly to the pre-training distribution.

5.1 Existence of Language Models' *Resistance*

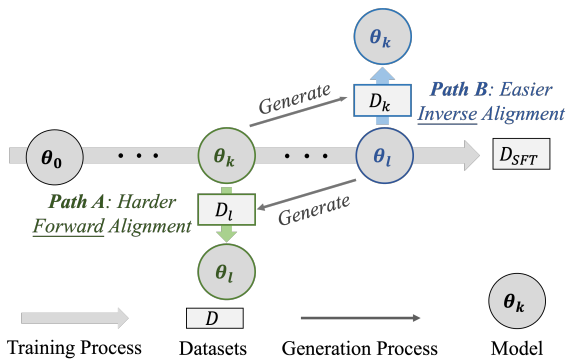

Figure 2: **Experiment pipeline for validating *resistance*.** We conceptualize *resistance* as: ***inverse alignment*** is easier than ***forward alignment***.

Experiment Design. We verify the existence of *resistance* by arguing that ***forward alignment*** is harder than ***inverse alignment*** for pre-trained models. Specifically, we first perform one epoch of SFT on a pre-trained LLM with parameters $\boldsymbol{\theta}_0$, saving the slices $\{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \ldots, \boldsymbol{\theta}_n\}$. Subsequently, without loss of generality, we collect the responses of slices $\boldsymbol{\theta}_k$ and $\boldsymbol{\theta}_l$ (where $k < l$) on hold-out prompts, forming datasets D_k and D_l. As shown in Figure 2, we define ***forward alignment* (*Path A*)** as the process of training $\boldsymbol{\theta}_k$ on D_l, and ***inverse alignment* (*Path B*)** as the process of training $\boldsymbol{\theta}_l$ on D_k.

Experiment Setup. We select different SFT datasets including Alpaca (Taori et al., 2023), TruthfulQA (Lin et al., 2022), and Beavertails (Ji et al., 2024c,b), which correspond respectively to the model's 3H principle (Askell et al., 2021). We divide them into three equal parts to obtain three corresponding slices $\{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3\}$. We consider Llama2-7B, Llama2-13B, and Llama3-8B (Touvron et al., 2023) as the base model $\boldsymbol{\theta}_0$.

Experiment Results. As shown in Table 1, the experimental results indicate that the training loss of ***inverse alignment*** consistently remains lower than that of ***forward alignment***, irrespective of the slice pair selection. This observation holds across all models and datasets in the experiments. All experimental results demonstrate that ***inverse alignment*** is easier than ***forward alignment*** across diverse models and datasets and validate the existence of *resistance*. To further verify the *resistance* in language models, we provide additional experimental details and results in the Appendix B.1.

5.2 Existence of *Rebound*

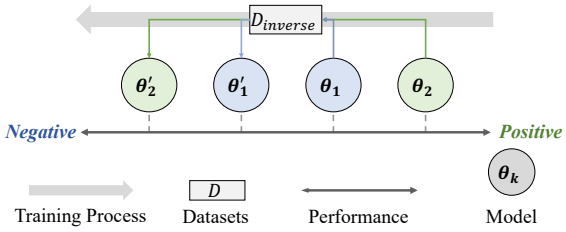

Figure 3: **Experiment pipeline for validating *rebound*.** We conceptualize *rebound* as: the more **positive** the post-trained models' performance, the more **negative** it becomes after inverse finetuning.

Experiment Design. We verify the existence of *rebound* by arguing that for post-trained models,

Datasets	Base Models	$\theta_2 \to \theta_1$ *vs.* $\theta_1 \to \theta_2$		$\theta_3 \to \theta_2$ *vs.* $\theta_2 \to \theta_3$		$\theta_3 \to \theta_1$ *vs.* $\theta_1 \to \theta_3$	
Alpaca	Llama2-7B	0.1589 ↓	0.2018 ↑	0.1953 ↓	0.2143 ↑	0.1666 ↓	0.2346 ↑
	Llama2-13B	0.1772 ↓	0.1958 ↑	0.2149 ↓	0.2408 ↑	0.1835 ↓	0.2345 ↑
	Llama3-8B	0.2540 ↓	0.2573 ↑	0.2268 ↓	0.3229 ↑	0.2341 ↓	0.2589 ↑
Truthful	Llama2-7B	0.1909 ↓	0.2069 ↑	0.1719 ↓	0.1721 ↑	0.2011 ↓	0.2542 ↑
	Llama2-13B	0.1704 ↓	0.1830 ↑	0.1544 ↓	0.1640 ↑	0.1825 ↓	0.2429 ↑
	Llama3-8B	0.2118 ↓	0.2256 ↑	0.2100 ↓	0.2173 ↑	0.2393 ↓	0.2898 ↑
Safe	Llama2-7B	0.2730 ↓	0.2809 ↑	0.2654 ↓	0.2691 ↑	0.2845 ↓	0.2883 ↑
	Llama2-13B	0.2419 ↓	0.2439 ↑	0.2320 ↓	0.2327 ↑	0.2464 ↓	0.2606 ↑
	Llama3-8B	0.2097 ↓	0.2156 ↑	0.2008 ↓	0.2427 ↑	0.2277 ↓	0.2709 ↑

Table 1: **Comparsion between *inverse alignment* and *forward alignment* training loss.** Under different model, task, and stage slicing settings, ***forward alignment*** is more challenging than ***inverse alignment***, *i.e.,* the loss is higher. This indicates that pre-trained models tend to maintain their original distribution.

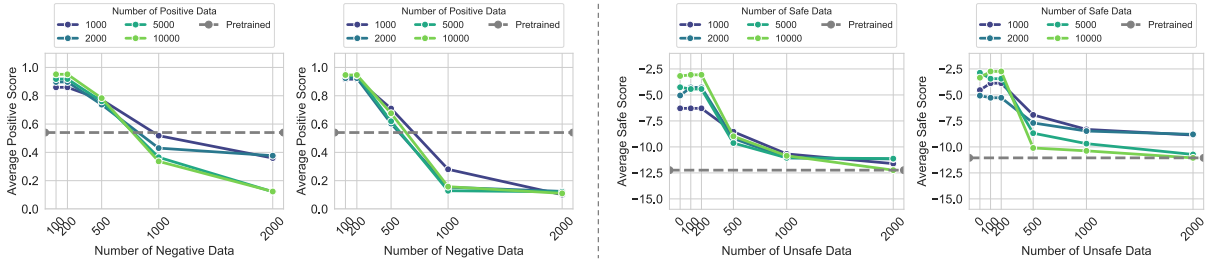

Figure 4: **Experimental results for validating the existence of *rebound* (left: IMDb, right: Beavertails).** The left part of each sub-figure is the performance of Gemma-2B while the right is Llama2-7B, respectively. Models trained with more **positive** data initially perform better but perform worse after fine-tuning with **negative** data.

the more **positive** the post-trained models' performance, the more **negative** it becomes after *inverse alignment*. We validate tasks involving two opposing characteristics (*e.g.* safe and unsafe). We first train slices $\{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, ..., \boldsymbol{\theta}_n\}$ based on a pre-trained model $\boldsymbol{\theta}_0$ using positive (*e.g.,* safe) data of various volumes. Then we perform inverse finetuning on these slices using negative data (*i.e.,* unsafe).

Tasks and Datasets. We select two tasks: positive generation and single-turn safe conversation. For the former, we use the data classified as positive or negative in the IMDb dataset (Maas et al., 2011). Referring to (Rafailov et al., 2024), we use the first 2-8 tokens of each complete text as a prompt for LLMs to generate the subsequent content. For the latter, we use the data classified as safe and unsafe in Beavertails (Ji et al., 2024c). We organized the positive sample sizes into {1000, 2000, 5000, 10000}, while the negative sample sizes, were divided into {100, 200, 500, 1000, 2000}.

Evaluation and Metrics. We collect the model's responses on the reserved test prompts. Then we use score models provided by existing research to complete the performance evaluation. For positive style generation, we refer to (Rafailov et al., 2024) and use the Sentiment Roberta model (Hartmann et al., 2023) to classify the responses, taking the proportion of all responses classified as positive as the model score. For single-turn safe dialogue, we use the cost model (Dai et al., 2024) to score the safety of each response, using the average score of all responses as the model score.

Experiment Results. We evaluate the *rebound* phenomenon on Llama2-7B (Touvron et al., 2023) and Gemma-2B (Team et al., 2024). The experimental results in Figure 4 show that, for models fine-tuned with a larger amount of positive sample data, their performance drops quicker under only a small amount of negative sample fine-tuning. Subsequently, the performance decline slows down and tends to stabilize. This result also confirms the previous conclusion: the initial rapid decline of model's performance is due to *rebound*, as the model is far from the pre-trained distribution; while the later stabilization of the countermeasure is due to *resistance*, as the model is already close to the pre-trained distribution.

To assess the generalizability of the *rebound* phenomenon, we perform additional ablation studies focusing on alignment algorithms, evaluation

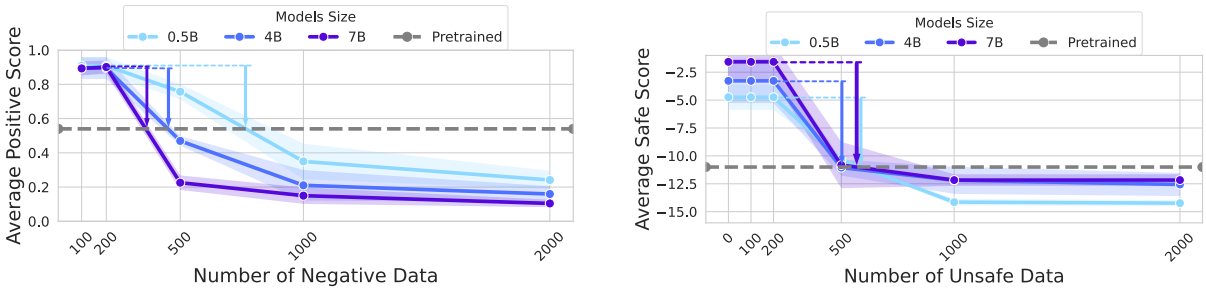

Figure 5: **Experimental results for validating *rebound* increases with model size (left: IMDb, right: Beavertails).** All single line covers positive data volume settings as Figure 4, with shadow denoting std. As the model size increases, the performance of the aligned model deteriorates more rapidly after fine-tuning with **negative** data.

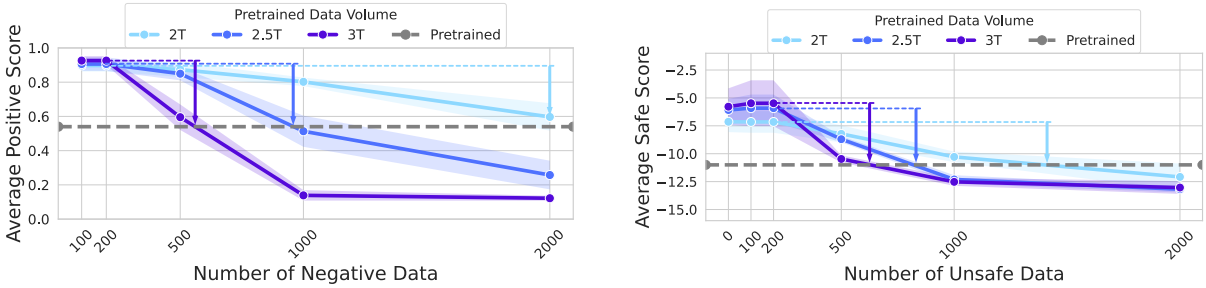

Figure 6: **Experimental results for validating *rebound* increases with pre-training data volume. (left: IMDb, right: Beavertails).** Each line covers positive data settings as Figure 4, with shadow denoting std. As pre-training data volume increases, aligned model performance deteriorates more rapidly after fine-tuning with **negative** data.

metrics, and fine-tuning directions. The results consistently validate the presence of the *rebound* phenomenon across language models. Further experimental details are available in Appendix B.2.

5.3 Internal Factor of *Rebound*

We conduct an in-depth analysis of the internal factors affecting the *rebound* phenomenon since it is crucial for the robustness of LLMs alignment. In particular, we investigate how model parameter scale and the amount of pre-training data influence *rebound*. Owing to space constraints, further experimental analyses are included in Appendix B.3.

***Rebound* Increases with Model Size.** To investigate how the *rebound* phenomenon varies with model size, we conducted experiments on Qwen models (Bai et al., 2023) with parameter scales of 0.5B, 4B, and 7B. The experimental results in Figure 5 show that as the model parameter size increases, the initial performance decline due to negative data fine-tuning is faster, while the subsequent decline is slower. This indicates that as the parameter size increases, there is an increase in *rebound* in response to both positive and negative data, further suggesting a positive correlation between model *elasticity* and parameter scale.

***Rebound* Increases with Pre-training Data Volume.** To verify that *rebound* increases with the growth of pre-training data, we vary pre-training slices (2.0T, 2.5T, and 3.0T) released by TinyLlama (Zhang et al., 2024a) and conduct the same experiment. As shown in Figure 6, when the pre-training data volume increases, the initial performance decline due to negative data fine-tuning is faster, while the subsequent decline is slower. It demonstrates that larger pre-training data volumes reinforce the *rebound* of LLMs, which is consistent with the inference proposed in Theorem 4.2.

6 Conclusion and Outlook

Our work uncovers a possible mechanism underlying the fragility of alignment: the *elasticity* of language models. We demonstrate that upon perturbation, the normalized compression rate of language models changes inversely with dataset size, making them more inclined to retain pre-training distributions while forgetting alignment distributions, thus resisting alignment. Through extensive experiments, we validate the universality of this *elasticity* effect, observing that the *elasticity* strengthens as model size and pre-training data volume scale up. Fundamentally, our findings support